SAMPLE

Solar System Missions: Pioneer

First to the Outer Planets

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Introduction

Before there was a Voyager, before any spacecraft had crossed the asteroid belt or felt the chill of the outer solar system, there was Pioneer. The name belonged, in the beginning, to a scattering of small probes in the late 1950s, and then to a patient family of solar observers, before it was given to two of the most consequential machines ever launched: Pioneer 10 and Pioneer 11, the first human-built objects to achieve escape velocity from the solar system. Between them, the two spacecraft crossed the asteroid belt, survived the radiation belts of Jupiter, mapped the approach to Saturn, carried the first physical message intended for other star systems, and, decades after their missions ended, presented science with one of its most instructive puzzles, a slowdown in their flight that took a generation to explain.

This book follows the Pioneer program from its earliest, often unsuccessful attempts in the years after Sputnik, through the solar-watchdog probes of the 1960s, to the design, launch, and planetary encounters of Pioneers 10 and 11. It retells the story of the gold-anodized plaque bolted to each craft, the debates it provoked on Earth, and the long, quiet years of flight beyond the planets. It closes with the anomaly that made the two aging probes scientifically valuable one last time, and with what their trajectories, their data, and their example left to the missions that followed. It is the story of machines sent first, and of everything that being first cost and taught.

Chapter 1: A Word That Chose Itself

A pioneer, in the old human sense, is someone who goes first, who clears the way for the people who come after, and it was almost inevitable that the word would migrate from covered wagons to rocket gantries. When the United States began reaching for space in the late 1950s, the name was attached to the very first attempts, and it stuck. The early Pioneers were not elegant. The first three, launched in 1958 under the joint supervision of the Air Force and the newly formed NASA, were built around the Able upper stages and meant partly as lunar probes, and the record they set was mostly a record of persistence. Pioneer 1 reached a then impressive altitude but fell short of the Moon; Pioneer 2 and Pioneer 3 met similar fates. Pioneer 4, in March 1959, finally achieved a proper lunar flyby trajectory, a small craft sailing past the Moon at a distance of some sixty thousand kilometers and proving that a spacecraft could be aimed across a substantial fraction of interplanetary space at all.

The most scientifically productive of the early series was Pioneer 5, launched in March 1960, which carried a magnetometer and became one of the first probes to operate in the space between Earth and Venus. Between roughly five and thirty-six million kilometers out, it confirmed that the Sun's magnetic field, carried by the solar wind, pervades interplanetary space, and it returned data until its transmitter, never designed for such distances, faded below the sensitivity of Earth's receivers. These were modest achievements by any later standard, but they established the pattern the name would carry: modest machines, doing something for the first time, so that larger machines could follow. Every subsequent craft to bear the word was, in one way or another, a test of ground that had never been tested before, whether the ground was a launch trajectory, a region of space, or an idea about what a spacecraft could survive.